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A palaeomagnetic study of Apennine thrusts, Italy: Monte Maiella and Monte Raparo

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Abstract

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Three separate structural blocks within the southern Apennines have been sampled for palaeomagnetic investigation to constrain their original separation and movement during mid-Tertiary deformation. The Mesozoic limestones are weakly magnetic, and the NRM intensity of all samples from Upper and Lower Cretaceous limestone from the Alburni platform and from Upper Cretaceous limestone at Monte Maiella were too low to yield results. Lower Cretaceous limestone at Monte Maiella contained a mean magnetisation (after structural correction) of $D = 326°$, $I = +42°$, $k = 44$, $N = 9$ (57° N, 263° E); and Cretaceous (?) limestone at Monte Raparo a mean of $D = 132°$, $I = -61°$, $k = 50$, $N = 19$ (54° N, 306° E). The Monte Maiella results, near the central part of the Apennine thrust-front, are compatible with a local, clockwise block-rotation during deformation, while Monte Raparo results may bear evidence of the major east-west thrust-motion during shortening in addition to anticlockwise block-rotations already reported from the southernmost Apennines.

Introduction

The Adriatic Promontory (Fig. 1), consisting of Bahamian type calcareous platforms and basins, remained an integral part of the African continent until the early Tertiary. However, palaeomagnetic evidence from Istria (Marton and Veljovic, 1983) and Gargano (Vandenberg, 1983) — both areas are considered to be autochthonous parts of the African foreland — suggests that Adria underwent a post-Oligocence anti-clockwise rotation with respect to Africa of about 17°, although its cause and exact areal extent is not clear.

In addition to microplate rotation, the originally wider Adriatic Promontory was shortened by mainly westerly thrusting in Miocene times as the European Sardinia–Calabria block collided with Adria due to the opening of the Ligurian Sea as a back-arc basin after the collision of the African and European plates (Channell, 1986). The thrusting produced the Apennine fold belt in Italy. Later, the suture between Sardinia–Calabria and Adria developed into a zone of extension and subsidence forming the Tyrrhenian Sea. Large clockwise rotation of thrusts in northern Sicily, and evidence of anticlockwise rotations in southern Italy, have been tied to this latter deformation (Channell et al., 1986).

The Apennines consist of a mosaic of thrusts and structural blocks, whose original position and motion during deformation are not always known in sufficient detail to allow restorations of sections across the fold belt. The suggestion that thrust rotations have occurred in its southern part (Incoronato, 1983; Incoronato et al., 1985; Manzoni, 1975, 1979) is of great interest, but many more studies are needed before local and regional effects can be separated and the age and causes of

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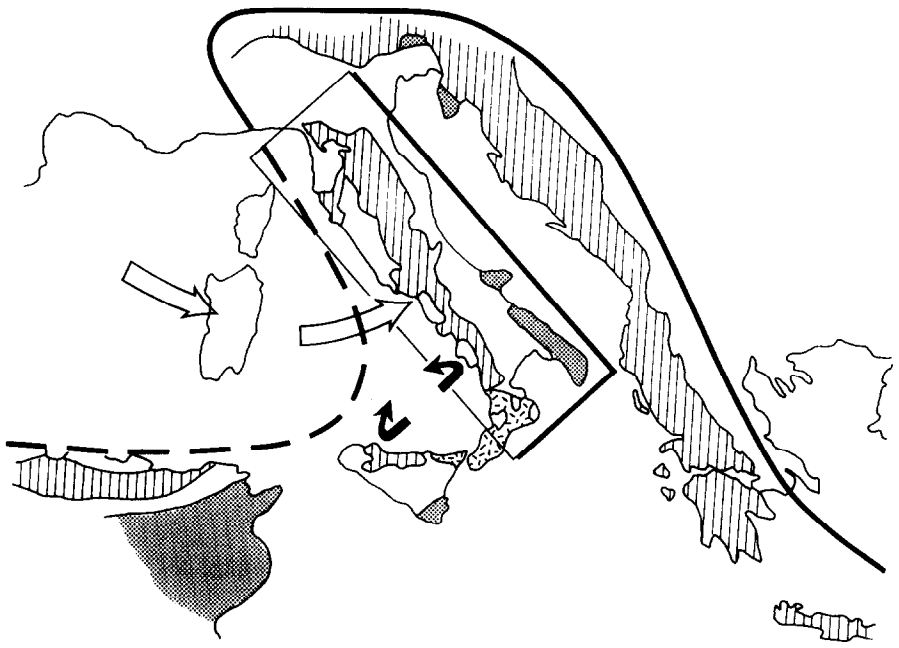


Fig. 1. Approximate extension of the Adriatic promontory of the African plate as defined by the occurrence of ophiolites and including South Tethyan continental margin facies (modified after Channell and Horvath, 1976). Arrows indicate general direction of thrust emplacement and rotations around steep axes. Legend and rectangle, see Fig. 2.

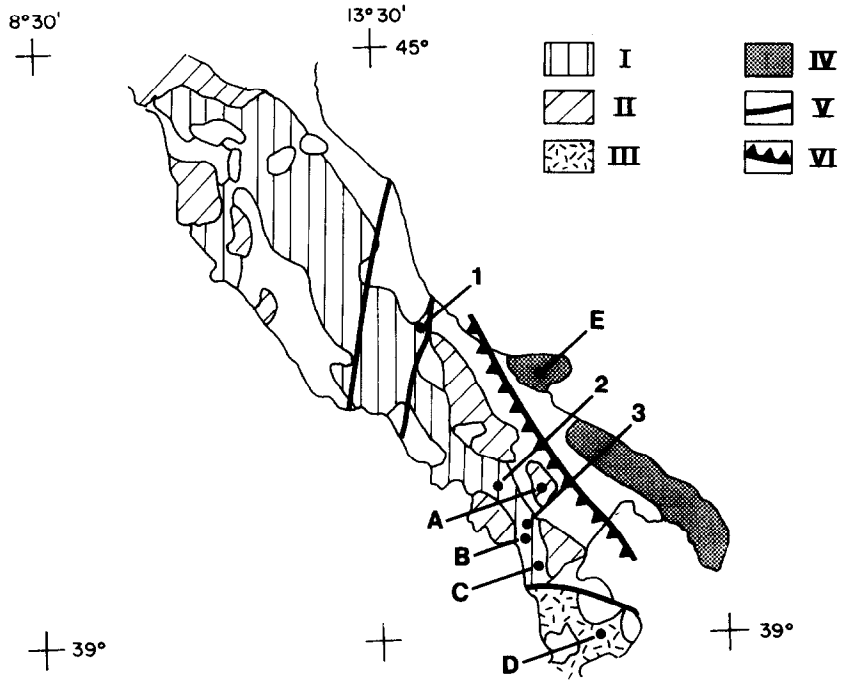


Fig. 2. Major stratigraphic and structural units in the southern Italy (modified after Channel and Horvath, 1976). Localities sampled for this study: *1* = Monte Maiella; *2* = the Alburno platform; *3* = Monte Raparo. Previous palaeomagnetic studies: *A* = Lagonegro upper nappe, Scisti Silicei (Incoronato et al., 1985) and Flysch Rosso (Incoronato, 1983); *B* = Lagonegro lower nappe (Incoronato et al., 1985); *C* = Monte Cerviero (Manzoni, 1975); *D* = Sila nappes (Manzoni, 1979), *E* = the Gargano peninsula (Lowrie, 1986). *Legend*: *I* = South Tethys continental margin (platforms and basins); *II* = oceanic remnants; *III* = metamorphosed units of uncertain original position; *IV* = African foreland; *V* = major tectonic lines; VI = Appennine thrust front.

the rotations determined. This paper presents the first part of a project which aims to constrain original separation and motion of thrusts in the southern Apennines (Fig. 2) using palaeomagnetic techniques. The results are compared with palaeopoles obtained from the Gargano peninsula, thereby eliminating the post-Oligocene microplate rotation of the Adriatic Promontory.

Sampling and laboratory methods

Three separate blocks of Mesozoic limestone were sampled in the southern Apennines (Fig. 2). Each site consists of three hand samples, individually oriented using a magnetic compass, from one sedimentary layer with vertical separation less than 1 m. Sites are commonly spaced 1–10 m apart.

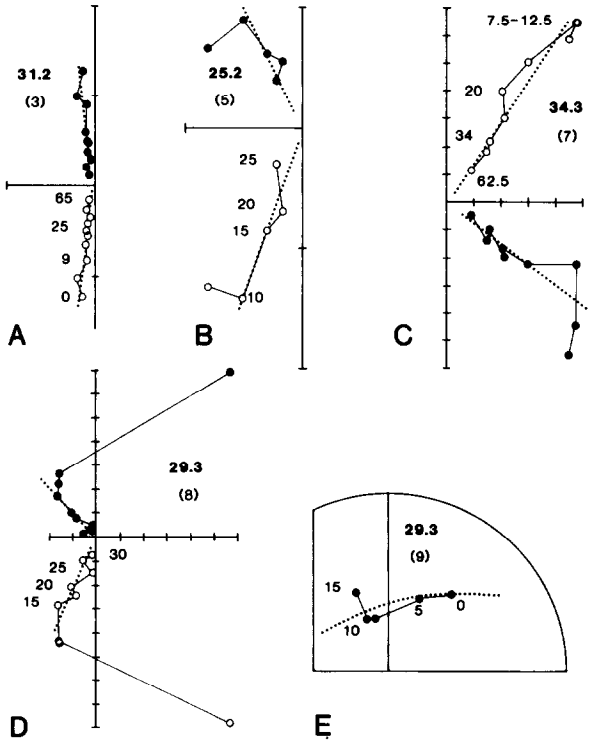


Fig. 3. Alternating field demagnetisation of specimens from Monte Maiella. Directions corrected for tilt. *A–D*. Orthogonal plots with demagnetisation steps in mT. Solid (open) circles represent projection of NRM vector tip onto horizontal (vertical) planes; scale divisions in 10^{-4} (A, D) or 10^{-5} (B, C) A/m. E. Wulff equal angle steronet with open circles on upper hemisphere. Dotted lines (great circle), approximate position of characteristic component (remagnetisation circle), mean angular deviation (Kirschvink, 1980) in brackets.

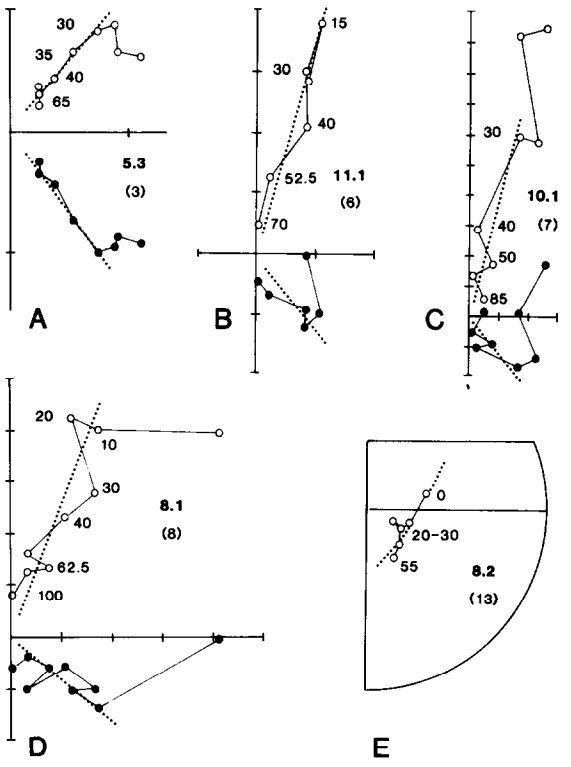


Fig. 4. Alternating field demagnetisation of specimens from Monte Raparo. Directions corrected for tilt. Denotations as in Fig. 3, scale divisions in 10^{-4} (A) or 10^{-5} (B–D) A/m.

(1) At Monte Maiella, on the western side of the small town of Fara (Italian Geological Survey Map of Rome, 1 : 100,000, Foglio 147: 42°05′15–30″N, 1°44′15″–45′00″E), 46 hand samples were collected, distributed between 4 sites in the Upper Cretaceous (Senonian–Barremian) unit, 7 in the uppermost part of the Lower Cretaceous (Barremian–Malm sup.) unit, and four sites approximately 300 m further down in the same unit.

(2) From the Alburno platform, 29 hand samples were collected at two separate localities: 5 sites in the Lower Cretaceous rocks (Cenomanian–Neocomian) on the northeast rim (Foglio 198 of Rome: 40°40′45″N, 2°57′45″E) and 5 sites in the Upper Cretaceous rocks (Senonian–Turonian) on the southwest rim (40°25′15″N, 2°57′50″E of Rome).

(3) At Monte Raparo, 11 sites or 28 hand samples were collected northwest of Castelsaraceno (Foglio 211 of Rome: 40°09′50″N, 3°31′50″E). Although the area is mapped as Early Cretaceous (Neoconian–Aptian), structural inves-

TABLE 1

Magnetic directions before and after simple tilt correction of beds, using strike and dip 223.2°E, +29.4° at sites 2–8, 243.2°E, +30.5° at sites 9–11, and 28.2°E, +41.9° at sites 25–34

Site	Sample *	In situ		Unfolded		N	k *
		D (°)	I (°)	D (°)	I (°)		
11	1	12.7	−87.4	149.7	−61.5	2 + 0	100
11	2	3.4	−70.2	116.3	−73.6	1 + 2	37
10	1	342.9	−82.6	150.0	−66.8	2 + 1	175
10	2	7.8	−67.9	106.9	−72.8	3 + 0	2000
10	3	94.7	−0.9	91.5	+14.5	0 + 3	24 [1]
9	2	3.7	−76.9	133.9	−70.1	2 + 1	501
8	1	189.1	−88.9	135.4	−64.9	3 + 0	500
8	2	93.4	−86.0	127.6	−62.5	1 + 2	130
8	3	109.4	−84.8	146.8	−55.8	3 + 0	133
7	1	121.0	−76.8	128.4	−54.6	1 + 0	–
7	2	111.5	−78.3	125.8	−54.4	1 + 2	265
7	3	73.9	−85.4	124.5	−63.0	1 + 0	–
6	2	50.4	−81.4	114.0	−63.2	3 + 0	182
6	3	83.9	−75.7	114.3	−54.7	1 + 1	2550
5	1	13.6	−87.2	127.1	−66.9	1 + 0	–
5	3	141.4	−58.8	138.6	−34.5	3 + 0	667
4	4	105.2	−84.0	131.2	−60.2	2 + 1	218
3	2	125.3	−73.4	129.7	−49.1	1 + 0	–
2	1	231.9	−76.3	165.5	−64.1	1 + 0	–
Mean		77.5	−85.3	132.0	−61.4	18 + 1	50 [3]
31	2	325.2	+1.5	333.3	+37.8	2 + 0	14 [2]
30	1	137.1	+14.1	138.6	−25.5	1 + 0	–
29	2	322.2	−9.7	325.3	+28.4	1 + 0	–
29	3	311.7	+11.0	319.7	+51.3	1 + 2	88
26	2	314.7	+11.2	324.3	+50.6	2 + 0	13
25	1	329.7	+3.8	339.7	+38.1	3 + 1	36
25	2	320.8	+12.9	333.8	+50.1	2 + 1	215
25	3	320.9	+6.9	330.8	+44.5	3 + 0	69
27	2	226.3	+36.5	192.8	+37.6	3 + 0	29
34	3	122.5	−14.8	126.5	−49.4	1 + 0	–
Mean		318.4	+4.3	326.0	+42.1	9 + 0	44

D = declination; I = inclination; N = number of specimens (samples), or characteristic components (sample directions) + remagnetisation circles (Bailey and Halls, 1984); k = Fisher (1953) precision parameter when based on characteristic components (sample directions), or as defined by Bailey and Halls (op. cit.) where remagnetisation circles are included.

* Superscripts: 1 = mean of poles to remagnetisation circles; 2 = mixed polarities; 3 = k in situ is 41.

tigation subsequent to sampling suggested that the limestones may be Triassic in age (R. Graham, pers. commun., 1986). The limestones contain no Foraminifera (S.S. Radford, pers. commun., 1988) and their age has therefore not been determined, but this uncertainty is not crucial to the discussion of results below.

The hand samples were drilled in the laboratory into 2.5 × 2.5 specimen cores. Specimens were treated using a Molspin Ltd. alternating field demagnetiser, a laboratory-built shielded furnace, and the natural remanent magnetisation (NRM) measured on a Molspin Ltd. spinner magnetometer. The specimens are weakly magnetic, but by multiple measurements it has been possible to detect directions with intensities as low as $1 \cdot 10^{-5}$ A/m (1 A/m = 10^{-3} emu/cm^3). The visual identification of components was complemented by

computer analysis, using either Principal Component Analysis (Kirschvink, 1980) or converging remagnetisation circles (Bailey and Halls, 1984). All accepted components (remagnetisation planes) have a mean angular deviation smaller than 8° (15°), examples are shown in Figs. 3 and 4. All sample directions (Table 1) are based upon fully demagnetised specimens; where possible, sample directions are based upon specimen components using either Fisher (1953) statistics, or, where remagnetisation circles are included, the combined analysis of Bailey and Halls (op. cit).

Palaeomagnetic results

Monte Maiella

Of 173 specimens drilled and measured from 46 samples, only 65 specimens contained a NRM $> 1 \cdot 10^{-5}$ A/m. Of these, 58 were subjected to

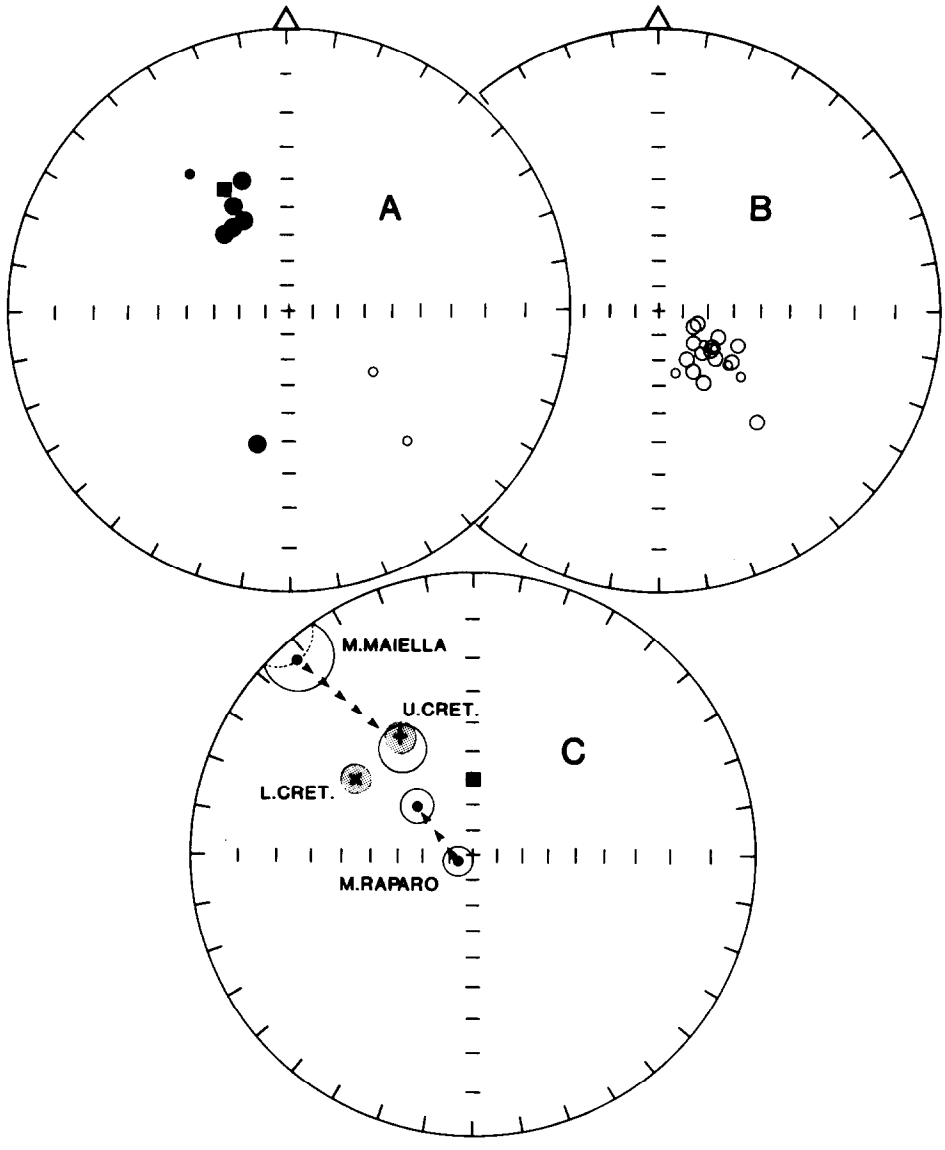


Fig. 5. Monte Maiella (A) and Monte Raparo (B) site directions after correction for tilt; Wulff equal angle stereonets, solid (open) symbols, lower (upper) hemisphere; square, mixed polarity site; small symbols, site direction based on single specimen. C. Site mean directions and circles of 95% confidence before and after tilt correction. Early (×) and Late (+) Cretaceous mean for the Gargano peninsula (Lowrie, 1986); square: recent Earth's field direction.

alternating field demagnetisation, but only 27 Lower Cretaceous specimens yielded a component (or remagnetisation circle) with angular deviation (Kirschvink, 1980) smaller than 8° (15°). Apart from a geologically insignificant overprint, the specimens contain a single, "characteristic" component (Fig. 3A–D), probably contained in magnetite as the coercivities do not exceed 80–100 mT. In a few cases, the characteristic component cannot be isolated, but its presence is indicated in the continous shift along a great circle (remagnetisation circle) during demagnetisation (Fig. 3E).

When individual specimen results are combined into sample directions (see Table 1), a further five specimens representing two samples were rejected because of internal scatter (k-value > 10). The Lower Cretaceous section contains directions of both normal and reverse polarity (Fig. 5A), but there is no other systematic change in direction between sites. Sample 27.2 was excluded from further consideration; in view of the presence of both polarities in the section, we suggest that this anomalous magnetic direction was acquired during a reversal. The Early Cretaceous mean direction for Monte Maiella (see Table 1) is based upon remaining sample directions, where reversed directions have been inverted to normal polarity. The magnetisation of the Monte Maiella limestones predates their folding, since the mean direction lies close to the Gargano directions only when the beds are unfolded to the horizontal around present strike (Fig. 5C).

The virtual geomagnetic pole position corresponding to the unfolded Monte Maiella direction is lat. 57° N, long. 263° E.

The Alburni Platform

Both Lower and Upper Cretaceous limestones at Monte Alburni were too weakly magnetic for palaeomagnetic techniques. The mean NRM of 69 specimens from 29 hand-samples is $7 \cdot 10^{-6}$ A/m, with scattered and unstable "directions" reflecting instrument noise rather than a palaeomagnetic signal and no further treatment or interpretation is possible.

Monte Raparo

The intensity of NRM is larger and also more variable than at Monte Maiella: from $2.5 \cdot 10^{-5}$ A/m for sites 1–4, to $5 \cdot 10^{-5}$ A/m for sites 5–11 with individual samples reaching $10\text{–}20 \cdot 10^{-5}$ A/m. Of 91 specimens drilled and measured from 28 hand samples (11 sites), 59 specimens were subjected to alternating field demagnetisation and 44 yielded a component (remagnetisation circle) with angular deviation smaller than 8° (15°). The weaker NRM intensity in the lower part of the section is also accompanied by greater instability with only three components isolated. All specimens contain, apart from a present Earth's or laboratory field overprint, a reversely magnetised characteristic component with no indication of subsidiary normal directions (Fig. 4). Since the Raparo direction is interpreted as secondary and indicating a complex deformation history (cf. below), six specimens were heated at 100, 200 and 300°C, and their NRM measured, to ensure that a truly single component has been isolated by alternating field demagnetisation. The specimens are fully demagnetised after being heated to 300°C, and the initially dispersed ($k = 4$) directions move to the unfolded mean at 200°C of $D = 134°$, $I = -60°$ ($k = 18$), in agreement with previous result (Table 1).

Sample directions, and the mean direction for Monte Raparo, are listed in Table 1 and plotted in Figs. 5B,C. The virtual geomagnetic pole position corresponding to the unfolded Monte Raparo direction is lat. 45° N, long. 306° E. When discussing the age and structural implications of these results, we will use the corresponding palaeopoles to avoid any error due to geographical separation of sampled sites.

Discussion

The Monte Maiella virtual geomagnetic polepoisition (vgp) is plotted in Fig. 6A together with Lower and Upper Cretaceous vgp's determined from the Gargano peninsula (Lowrie, 1986). The Monte Maiella result is based on Lower Cretaceous rocks sampled from the higher part of a section indicated as Barremian to Malm sup. on the 1 : 25,000 geological map of Italy (Foglio 147). The Maiella vgp is, however, significantly displaced at the 95% confidence level from the Early Cretaceous Gargano position and coincides instead with the Late Cretaceous position. In view of the presence of both normal and reversely

magnetised samples, it is reasonable to assume that the magnetisation is primary and contemporaneous with the deposition and diagenesis of the limestone. It is then necessary to rotate the Maiella vgp, which is most easily done using a vertical axis through the sampled locality. A 24° anticlockwise rotation brings the Monte Maiella vgp closest to the Early Cretaceous Gargano direction as given by their maximum combined vector sum. Unfortunately, the Gargano results do not include lists of individual directions corrected for tilt and it is not possible to undertake a proper error analysis (Watson, 1956). Instead, a conservative estimate of the overlap of circles of confidence in Fig. 6A suggests an error of ±15°. In other words, the Monte Maiella result can be explained by a 24 ± 15° clockwise rotation around a vertical axis, and there is no positive evidence to

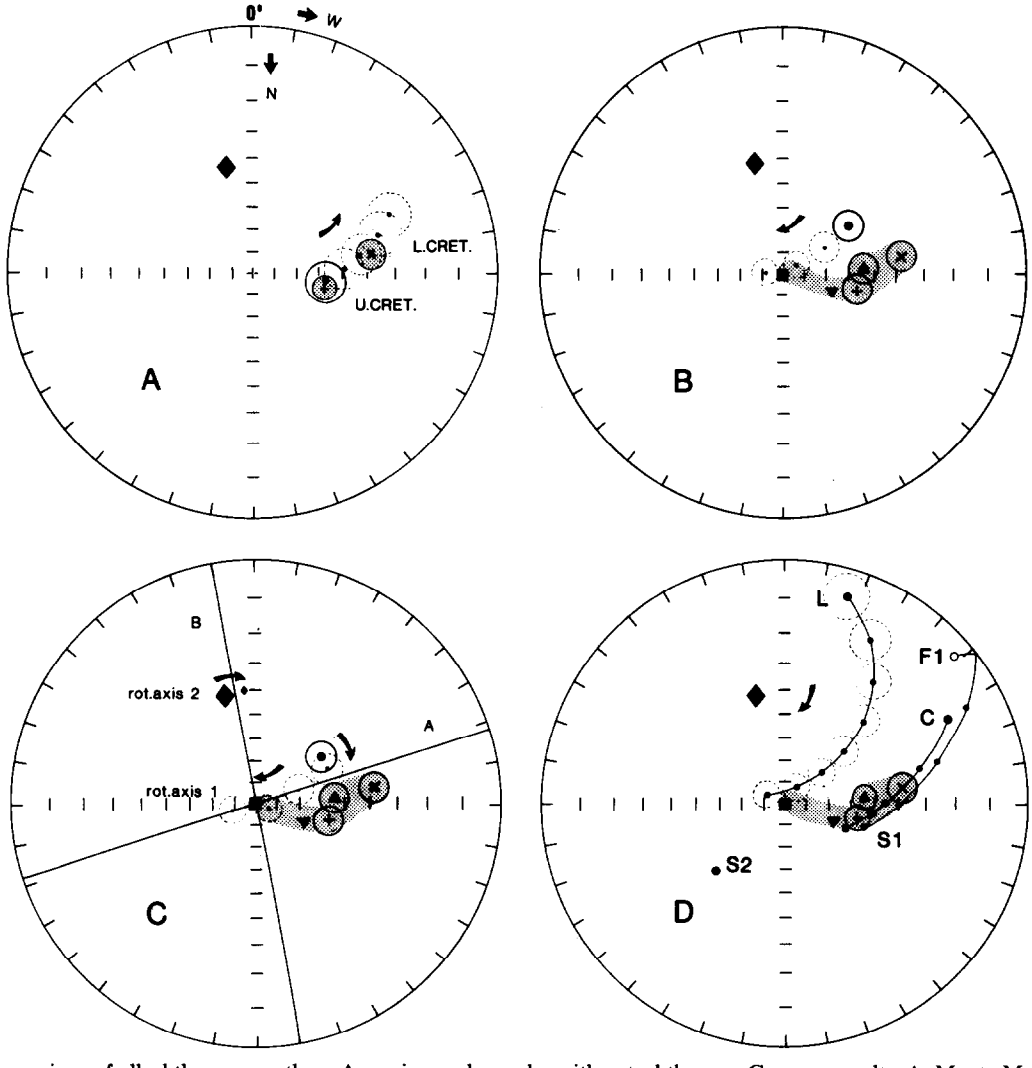


Fig. 6. Comparison of allochthonous southern Apennines palaeopoles with autochthonous Gargano results. A. Monte Maiella, small dots and dashed circles show anticlockwise rotation of vgp in 10° steps around a local, vertical axis (◆). B. Monte Raparo, with clockwise rotation in 20° steps. C. Clockwise 10° rotation of Monte Raparo and its vgp around rotation axis *1* (see text), followed by rotation as in B. D. Previously published palaeopoles: *S1*, *S2* = Scisti Scilicei, Lagonegro upper nappe; *L* = Scisti Silicei, Lagonegro lower nappe, steep component; *C* = Monte Cerviero; *F1* = Flysch Rosso; with rotation in 20° steps around vertical axis of *L*, *C* and *F1*. Wulff equal angle stereonets, where circumference equals the equator; circles of 95% confidence; shaded area, tentative apw path for Adria: ▲ = mean of Late Permian to Middle Jurassic African results (Vandenberg, 1979); ×, + = Early, Late Cretaceous, Gargano and ▼ Oligocene, Umbria (Lowrie, 1986); ■ = Recent Earth's field.

suggest that Monte Maiella did not originate close to its present position.

The Monte Raparo vgp is plotted in Fig. 6B together with a tentative Triassic–Recent apparent polar wander (apw) path for Adria (shaded). The path is based upon Gargano Cretaceous results, an Oligocene result from Umbria (Lowrie, op. cit.), as well as the mean calculated from Late Permain to Middle Jurassic African results (Vanderberg, 1979) and rotated by 17° around a pole at Malta to eliminate the difference due to assumed microplate rotation of Adria. No post-Eocence palaeo-poles have been reported from mainland Italy, and I assume a simple shift between the Oligocene and a Recent Earth's field averaging present secular variation. The Monte Raparo result is obtained from Triassic or Lower Cretaceous limestones but its vgp, whether based on in situ or tilt-corrected mean direction, is significantly different at the 95% confidence level from any section of the apparent polar wander path. The tilt-correction brings the vgp closer to the path (cf. Fig. 5C), but, in contrast to Monte Maiella, it is still not possible to bring the vgp to coincide with any but the most recent section of the path by a simple rotation around a vertical axis through the sampled site (Fig. 6B).

To bring the Monte Raparo vgp to agree with the apw path, it is necessary to use either an inclined, local axis or an axis displaced from the sampled area which would be affected as well. A rotation-axis can be found by a simple geometrical construction, illustrated in Fig. 6C, where it is assumed that the limestones were deposited and magnetised in the Triassic, and that the displacement of the vgp is due (in part) to motion of Monte Raparo itself along the Earth's surface. The vgp can be rotated to the Triassic position using any axis which lies on the great-circle plane *A* that perpendicularly bisects the great-circle arc connecting the two points. The movement of Monte Raparo is constrained by the geological evidence for a mainly easterly mid-Tertiary Apennine thrust-motion: if, for example, Monte Raparo (41°N, 16°E) originated as far as 10° to the east in the present Ligurian Sea, its motion occurred around an axis on great-circle plane *B*. The intersection between these two circles gives the axis around which both can be rotated. It is, however, not possible to rotate both by the same amount: a 10° rotation will bring Monte Raparo to 50°E, but is insufficient to bring the vgp to agree with the Early Mesozoic position as the two remain significantly different at the 95% confidence level. Thus, Apennine thrust-motion cannot explain the displacement of the vgp if the magnetisation is assumed to be primary or coeval with the formation of the rock. (If the magnetisation is Cretaceous, great circle *A* and thus rotation axis *1* would shift somewhat, but the conclusion remains). A similar geometrical argument can be used to find a local, inclined axis to rotate a primary magnetisation to its Early Mesozoic position and the tilted beds to the horizontal; but, yet again, it is easily shown that it is not possible to rotate both by the same amount. In conclusion, a primary magnetisation cannot be reconciled with the constraint upon thrust-motion or the requirement that beds were magnetised in the horizontal position.

Therefore, the magnetisation is secondary, and the most likely remagnetisation agent is a slight increase in temperature and/or the introduction of precipitating fluids in connection with the mid-Tertiary deformation. Although block-rotation around a vertical axis is not by itself sufficient to bring the vgp to a Tertiary position (Fig. 6B), it is suggestive (Fig. 6C) that the 10° clockwise rotation around axis *1*, discussed above, combined with a 40° clockwise rotation around a vertical axis (*2*) will bring the vgp to the mid-to-late Tertiary section of the apw path. In other words, the palaeomagnetic results could be interpreted as indicating a Tertiary magnetisation, acquired by the limestones before the main stages of a deformation, which included a major westerly thrust-motion of Monte Raparo accompanied or succeeded by its clockwise 40° rotation around a vertical axis.

On the basis of the Monte Raparo direction alone, it would not be possible to rule out the alternative explanation that the offset of the vgp from the apparent polar wander path was due to magnetisation at an intermediate deformation stage with beds tilted around an axis other than present strike (cf. Fig. 5C). It is therefore im-

portant that the Jurassic Scisti Silicei Formation of the Lagonegro lower nappe (Incoronato et al., 1985, sampled close to Monte Raparo — *B* in Fig. 2) contains a magnetisation, dismissed by the authors as inconsistent and of unknown origin, which is remarkably similar to that of Monte Raparo in its negative polarity and steep inclination, although its declination suggests that an even larger anticlockwise rotation (Fig. 6D; as before an element of rotation around axis *1*, or easterly thrust-motion, is also desirable, although not included in the figure).

Results reported by Incoronato et al. (op. cit.) from the Jurassic Scisti Silicei, Lagonegro upper nappe, and by Incoronato (1983) from the Oligocene–Upper Cretaceous Red Flysch, both sampled farther northeast of Monte Raparo (*A* in Fig. 2), are more difficult to assess. Two directions (*S1* and *S2*, Fig. 6D), often residing in the same specimen, were obtained from the Scisti Silicei; *S1* is regarded by the authors as a Cretaceous overprint and it is, therefore, incorrect to interpret *S2* as evidence of Tertiary thrust-rotation as the one cannot be rotated without affecting the other. The Upper Cretaceous Red Flysch contains one component (*F1*) which could be interpreted as evidence of anticlockwise rotation (Fig. 6D); a second component, which appears to lie in a similar relative position to *F1* as *S2* does to *S1*, is also indicated by great circle motion during demagnetisation but not discussed by the author. Thus, these sediments contain enigmatic components which cannot be rotated to agree with the apparent polar wander path, and a final estimate of thrust-motion must wait until the provenance of these components is determined. It seems uncontested, however, that anticlockwise rotations local, around steep axes occurred in Calabria (*C–D* Fig. 2) as Manzoni (1975, 1979) measured a rotation of 73° with respect to Africa (50° with respect to Gargano, Fig. 6D) of Upper Cretaceous limburgites, and of 90° with respect to the Apennines of Lower Permian rocks from the Sila nappes.

Conclusions

(1) Palaeomagnetic results from the Lower Cretaceous limestones at Monte Maiella have yielded a magnetisation of normal and reverse polarity with a mean direction of $D = 326.0°$, $I = +42.1°$ ($k = 44$, $N = 9$) which corresponds to a virtual geomagnetic pole position at 57° N and 263° E.

(2) The Monte Maiella palaeopole is significantly different at the 95% confidence level from the Early Cretaceous palaeopole based upon results from the autochthonous Gargano peninsula. The offset is most easily explained by a $24 \pm 15°$ clockwise rotation of the Maiella block around a vertical axis during the Mid-Tertiary Apennine deformation.

(3) The mean magnetic direction of Triassic or Lower Cretaceous limestone at Monte Raparo is $D = 132.0°$, $I = -61.4°$ ($k = 50$, $N = 19$) which corresponds to a virtual geomagnetic pole position at 54° N, 306° E.

(4) The Monte Raparo palaeopole is significantly different at the 95% confidence level from measured or inferred Mesozoic and younger results from autochthonous Italy and Africa. The Monte Raparo magnetisation is secondary — and probably acquired in connection with the Apennine deformation — since it cannot be brought to agree with Mesozoic results given the geological constraints upon thrust-motions.

(5) Although too far-reaching conclusions should not be drawn from as yet limited results, we suggest that the Monte Raparo magnetisation was acquired in an early stage of deformation, before the beds were folded, and that the comparison of the paleopole with its expected position suggests that Monte Raparo has undergone a large E–W latitudinal thrust motion as well as a substantial (40°) anticlockwise rotation around a vertical axis.

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